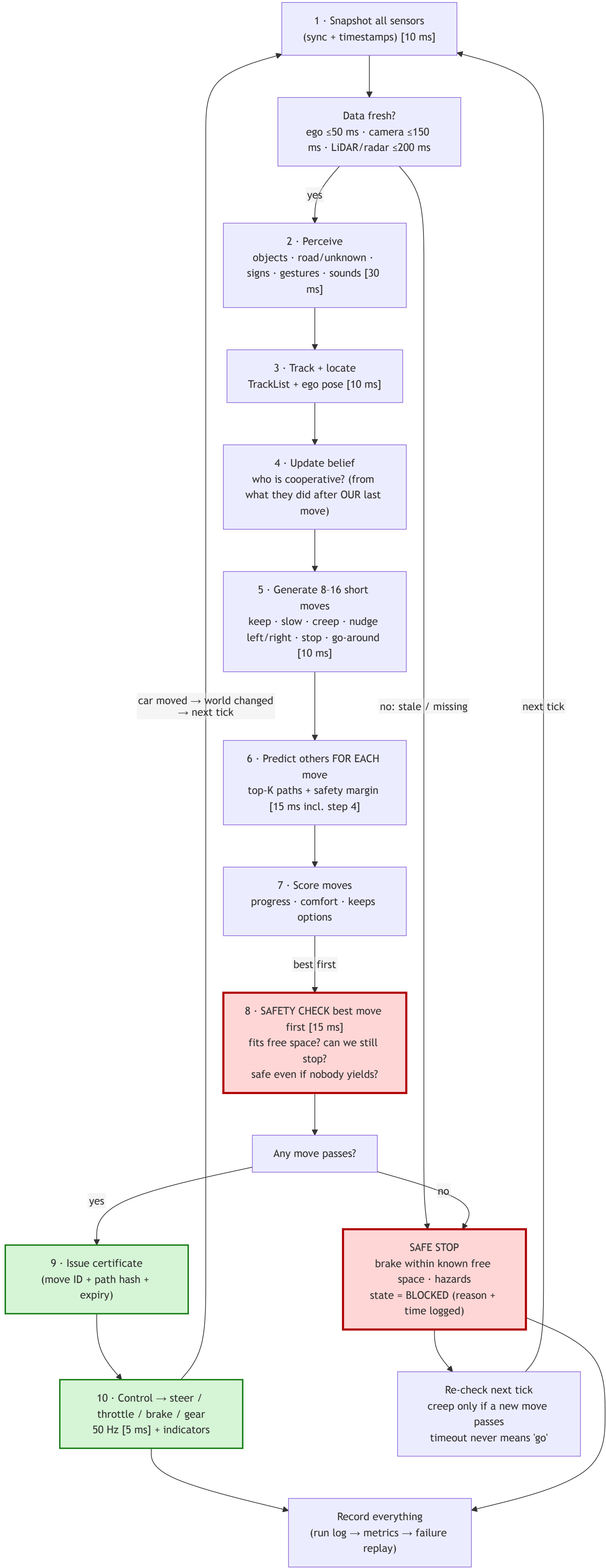


D2 · One decision, step by step (every 100 ms)

Green = the car moves. Red = safety check / safe stop. Brackets = target time budget (today: 736 ms per step).



D2 · One decision, step by step (every 100 ms) — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

BOXES (grouped exactly as drawn):

- EVERY 100 ms (10 Hz planning tick) — timing budget in brackets
 - [s1] 1 · Snapshot all sensors (sync + timestamps) [10 ms]
 - [s0] Data fresh? ego ≤ 50 ms · camera ≤ 150 ms · LiDAR/radar ≤ 200 ms
 - [s2] 2 · Perceive objects · road/unknown · signs · gestures · sounds [30 ms]
 - [s3] 3 · Track + locate TrackList + ego pose [10 ms]
 - [s4] 4 · Update belief who is cooperative? (from what they did after OUR last move)
 - [s5] 5 · Generate 8–16 short moves keep · slow · creep · nudge left/right · stop · go-around [10 ms]
 - [s6] 6 · Predict others FOR EACH move top-K paths + safety margin [15 ms incl. step 4]
 - [s7] 7 · Score moves progress · comfort · keeps options
 - [s8] 8 · SAFETY CHECK best move first [15 ms] fits free space? can we still stop? safe even if nobody yields?
 - [s9] Any move passes?
 - [s10] 9 · Issue certificate (move ID + path hash + expiry)
 - [s11] 10 · Control → steer / throttle / brake / gear 50 Hz [5 ms] + indicators
 - [s12] SAFE STOP brake within known free space · hazards state = BLOCKED (reason + time logged)
 - [s13] Re-check next tick creep only if a new move passes timeout never means 'go'
 - [s14] Record everything (run log → metrics → failure replay)

CONNECTIONS (18 arrows, from → to : what flows):

1 · Snapshot all sensors (sync + timestamps) [10 ms] → Data fresh? ego ≤ 50 ms · camera ≤ 150 ms · LiDAR/radar ≤ 200 ms
Data fresh? ego ≤ 50 ms · camera ≤ 150 ms · LiDAR/radar ≤ 200 ms → 2 · Perceive objects · road/unknown · signs · gestures · sounds [30 ms] : yes
Data fresh? ego ≤ 50 ms · camera ≤ 150 ms · LiDAR/radar ≤ 200 ms → SAFE STOP brake within known free space · hazards state = BLOCKED (reason + time logged) : no: stale / missing
2 · Perceive objects · road/unknown · signs · gestures · sounds [30 ms] → 3 · Track + locate TrackList + ego pose [10 ms]
3 · Track + locate TrackList + ego pose [10 ms] → 4 · Update belief who is cooperative? (from what they did after OUR last move)
4 · Update belief who is cooperative? (from what they did after OUR last move) → 5 · Generate 8–16 short moves keep · slow · creep · nudge left/right · stop · go-around [10 ms]
5 · Generate 8–16 short moves keep · slow · creep · nudge left/right · stop · go-around [10 ms] → 6 · Predict others FOR EACH move top-K paths + safety margin [15 ms incl. step 4]
6 · Predict others FOR EACH move top-K paths + safety margin [15 ms incl. step 4] → 7 · Score moves progress · comfort · keeps options
7 · Score moves progress · comfort · keeps options → 8 · SAFETY CHECK best move first [15 ms] fits free space? can we still stop? safe even if nobody yields? : best first
8 · SAFETY CHECK best move first [15 ms] fits free space? can we still stop? safe even if nobody yields? → Any move passes?
Any move passes? → 9 · Issue certificate (move ID + path hash + expiry) : yes
Any move passes? → SAFE STOP brake within known free space · hazards state = BLOCKED (reason + time logged) : no
9 · Issue certificate (move ID + path hash + expiry) → 10 · Control → steer / throttle / brake / gear 50 Hz [5 ms] + indicators
SAFE STOP brake within known free space · hazards state = BLOCKED (reason + time logged) → Re-check next tick creep only if a new move passes timeout never means 'go'
Re-check next tick creep only if a new move passes timeout never means 'go' → 1 · Snapshot all sensors (sync + timestamps) [10 ms] : next tick
10 · Control → steer / throttle / brake / gear 50 Hz [5 ms] + indicators → Record everything (run log → metrics → failure replay)
SAFE STOP brake within known free space · hazards state = BLOCKED (reason + time logged) → Record everything (run log → metrics → failure replay)
10 · Control → steer / throttle / brake / gear 50 Hz [5 ms] + indicators → 1 · Snapshot all sensors (sync + timestamps) [10 ms] : car moved → world changed → next tick

Mermaid source (paste into mermaid.live to re-render)

flowchart TB

```
s1["1 · Snapshot all sensors<br/>(sync + timestamps) [10 ms]"]
s0["Data fresh?<br/>ego ≤50 ms · camera ≤150 ms · LiDAR/radar ≤200 ms"]
s2["2 · Perceive<br/>objects · road/unknown · signs · gestures · sounds [30 ms]"]
s3["3 · Track + locate<br/>TrackList + ego pose [10 ms]"]
s4["4 · Update belief<br/>who is cooperative? (from what they did after OUR last move)"]
s5["5 · Generate 8–16 short moves<br/>keep · slow · creep · nudge left/right · stop · go-around [10 ms]"]
s6["6 · Predict others FOR EACH move<br/>top-K paths + safety margin [15 ms incl. step 4]"]
```

```

s7["7 · Score moves<br/>progress · comfort · keeps options"]
s8["8 · SAFETY CHECK best move first [15 ms]<br/>fits free space? can we still stop?<br/>safe even
if nobody yields?"]
s9["Any move passes?"]
s10["9 · Issue certificate<br/>(move ID + path hash + expiry)"]
s11["10 · Control → steer / throttle / brake / gear<br/>50 Hz [5 ms] + indicators"]
s12["SAFE STOP<br/>brake within known free space · hazards<br/>state = BLOCKED (reason + time
logged)"]
s13["Re-check next tick<br/>creep only if a new move passes<br/>timeout never means 'go'"]
s14["Record everything<br/>(run log → metrics → failure replay)"]
s1 --> s0
s0 -->|"yes"| s2
s0 -->|"no: stale / missing"| s12
s2 --> s3
s3 --> s4
s4 --> s5
s5 --> s6
s6 --> s7
s7 -->|"best first"| s8
s8 --> s9
s9 -->|"yes"| s10
s9 -->|"no"| s12
s10 --> s11
s12 --> s13
s13 -->|"next tick"| s1
s11 --> s14
s12 --> s14
s11 -->|"car moved → world changed → next tick"| s1
classDef safety fill:#ffd6d6,stroke:#b00000,stroke-width:3px
class s8,s12 safety
classDef go fill:#d6f5d6,stroke:#1a7f1a,stroke-width:2px
class s10,s11 go

```